

JUnit Basic Exercises

Exercise 1: Setting Up JUnit:

Pom.xml:

```
m pom.xml (JUnit_Demo) x
2      <project xmlns="http://maven.apache.org/POM
32      <dependencies>
43  @  <dependency>
44      <groupId>junit</groupId>
45      <artifactId>junit</artifactId>
46      <version>4.13.2</version>
47      <scope>test</scope>
48      </dependency>
49  </dependencies>
```

Calculator Class:

```
public class Calculator { no usages new *
    public int add(int a, int b) { new *
        return a + b;
    }
    public int subtract(int a, int b) { no usages new *
        return a - b;
    }
}
```

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Exercise 2: Writing Basic JUnit Tests

Test Class:

```
import org.junit.Test;
import static org.junit.Assert.assertEquals;

public class CalculatorTest { new *
    @Test new *
    public void testAdd() {
        Calculator obj = new Calculator();
        int result = obj.add( a: 10, b: 20);
        assertEquals( expected: 30, result);
    }

    @Test new *
    public void testSubtract() {
        Calculator obj = new Calculator();
        int result = obj.subtract( a: 20, b: 5);
        assertEquals( expected: 15, result);
    }
}
```

Output:

✓ CalculatorTest	13 ms	✓ 2 tests passed 2 tests total, 13 ms
✓ testAdd	13 ms	"C:\Program Files\Java\jdk-25\bin\java.exe" ...
✓ testSubtract	0 ms	Process finished with exit code 0

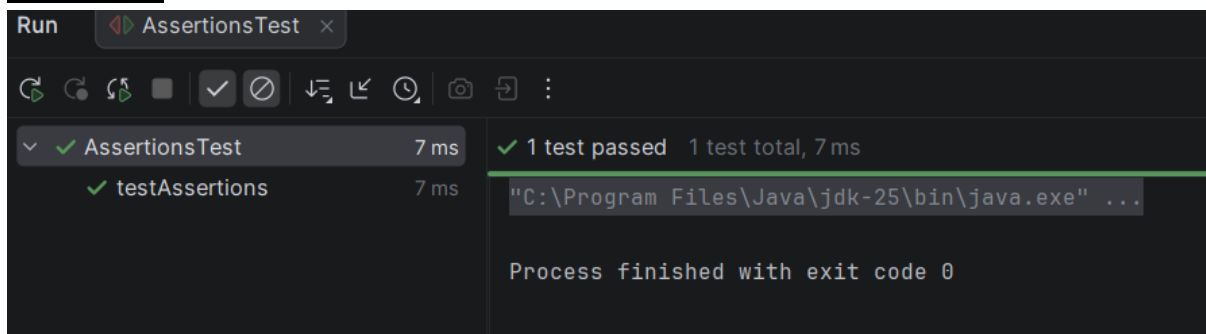
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Exercise 3: Assertions in JUnit

AssertionsTest Class:

```
import org.junit.Test;
import static org.junit.Assert.*;
public class AssertionsTest {
    @Test
    public void testAssertions() {
        assertEquals( expected: 5, actual: 2 + 3);
        assertTrue( condition: 10 > 5);
        assertFalse( condition: 5 > 10);
        assertNull( object: null);
        assertNotNull(new Object());
    }
}
```

Output:



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Exercise 4: Arrange-Act-Assert (AAA) Pattern, Test Fixtures, Setup and Teardown Methods in JUnit

CalculatorAAATest Class:

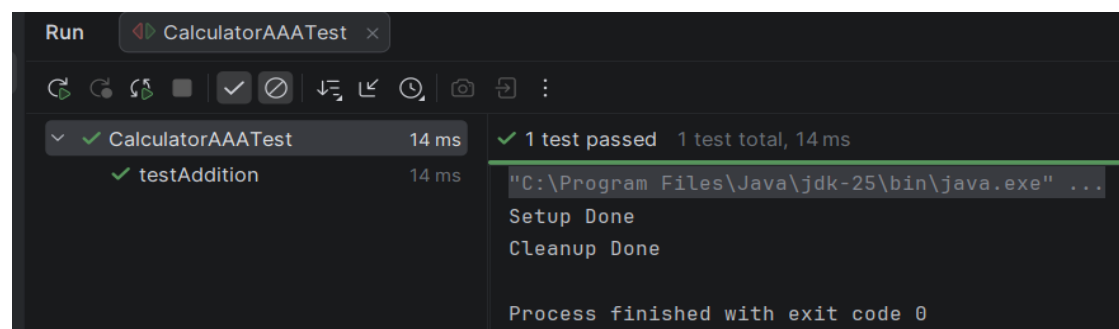
```
import org.junit.Before;
import org.junit.After;
import org.junit.Test;
import static org.junit.Assert.assertEquals;

public class CalculatorAAATest { new *
    Calculator calculator; 3 usages
    @Before new *
    public void setUp() {
        calculator = new Calculator();
        System.out.println("Setup Done");
    }

    @After new *
    public void tearDown() {
        calculator = null;
        System.out.println("Cleanup Done");
    }

    @Test new *
    public void testAddition() {
        // Arrange
        int first = 15;
        int second = 10;
        // Act
        int result = calculator.add(first, second);
        // Assert
        assertEquals("expected: 25, result);
    }
}
```

Output:



The screenshot shows the IDE's Run window for the `CalculatorAAATest` class. The test results table indicates that the `testAddition` test passed successfully in 14 ms. The console output shows the messages "Setup Done" and "Cleanup Done", confirming that the setup and teardown methods were executed as expected. The process finished with an exit code of 0.

Test Name	Duration	Status
CalculatorAAATest	14 ms	Passed
testAddition	14 ms	Passed

1 test passed 1 test total, 14 ms

"C:\Program Files\Java\jdk-25\bin\java.exe" ...

Setup Done

Cleanup Done

Process finished with exit code 0